Analysis of Gene Expression in TCGA-BRCA Cohort

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1 Introduction

The data for this project were gathered from The Cancer Genome Atlas (TCGA) program's Breast Invasive Carcinoma (TCGA-BRCA) cohort [3]. The dataset includes RNA-seq count data and associated metadata from breast cancer patients. The metadata contain patient demographics, tumor characteristics, treatment information, survival outcomes, and patient sample details. Samples were collected from tumor tissue and from matched normal tissue from each patient (usually the patient's blood). The gene expression dataset contains RNA-seq count data for 60,660 genes across 1,231 samples.

Estrogen Receptor 1 (ESR1) is a widely studied gene in breast cancer that plays a major role in estrogen signaling. Research shows that tumors with high expression are often hormone receptor-positive and are more likely to respond to endocrine therapy than tumors that show low ESR1 expression [2]. Likewise, Erb-B2 Receptor Tyrosine Kinase 2 (ERBB2) is another widely studied gene in breast cancer research. High expression of ERBB2 can stimulate cell proliferation and tumor growth, yielding a more aggressive form of disease. As such, high ERBB2 expression is associated with adverse prognosis and increased risk of disease progression [1]. Together, ESR1 and ERBB2 are two prognostic biomarkers that frequently determine treatment in breast cancer patients.

2 Methods

All analyses were conducted in R. Data import, wrangling, summary statistic generation, and most visualizations were conducted using the tidyverse package (version 2.0.0). The gene expression heatmap was generated using ComplexHeatmap (version 2.28.0) with circlize (version 0.4.18) for color-mapping. Finally, the Uniform Manifold Approximation and Projection (UMAP) was generated using the uwot package (version 0.2.5).

3 Results

3.1 Summary Statistics

Table 1: Summary statistics for ESR1 expression across samples.

mean	median	sd	min	max
31442.22	17561	38261.25	14	270977

The above summary statistics (Table 1) indicate high variability in ESR1 expression level across samples. First, the standard deviation (38,261) is greater than the mean, which indicates a wide spread of expression values around the mean. Also, the max (270,977) is substantially greater than the minimum (14), which can further indicate high variability. The max is also substantially higher than both the mean and median, which indicates the presence of a subset of tumors with extremely high ESR1 expression. Further supporting this observation, the mean expression value is about 1.8x greater than the median, which suggests a positively skewed distribution in which a relatively small number of high-expression samples pull the mean upward.

Table 2: Distribution of samples by pathologic stage.

ajcc_pathologic_stage	n	Percent
Stage 0	4	0.4
Stage I	98	8.8
Stage IA	93	8.4
Stage IB	6	0.5
Stage II	7	0.6
Stage IIA	374	33.6
Stage IIB	267	24.0
Stage IIIA	155	13.9
Stage IIIB	27	2.4
Stage IIIC	62	5.6
Stage IV	20	1.8

The above statistics (Table 2) show that about 72% of all samples were classified between Stage IIA and Stage IIIA, with Stage IIA representing the largest group (33.6%). About 19% of samples were classified between Stage 0 and Stage II (i.e., before Stage IIA), with 17.2% in Stages I and IA. As such, about 10% of samples were classified between Stage IIIB and Stage IV, the smallest proportion (1.8%) being Stage IV. Overall, the data suggest that the samples were mostly intermediate-stage tumors, with about 10% being more advanced-stage.

Table 3: Summary statistics for age at diagnosis.

mean	median	sd	min	max
58.9	58.8	13.2	26.9	90

The above statistics (Table 3) show the average age at diagnosis was about 58.9 years, with a standard deviation of approximately 13.2 years. The median age (58.8) is almost equal to the mean, indicating a relatively symmetric age distribution. Further, the minimum age was about 26.9 years, and the max about 90.0 years, which highlights that there were both relatively young and elderly patients at diagnosis.

3.2 Histogram, Scatter Plot, and Violin Plot

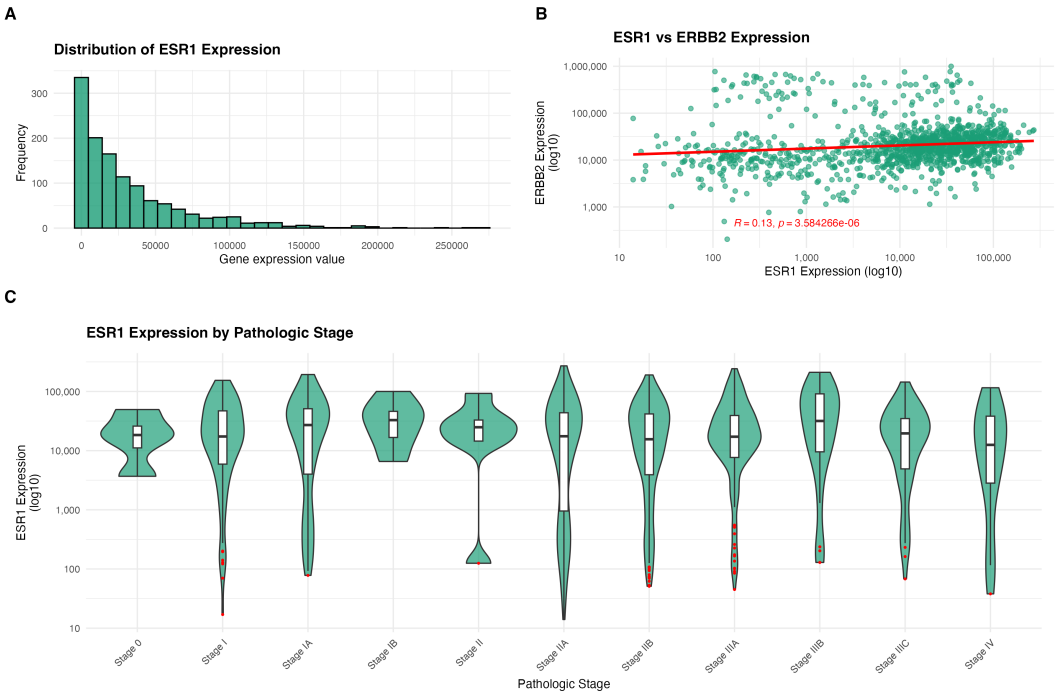


Figure 1: A. "Distribution of ESR1 Expression" histogram, **B.** "ESR1 versus ERBB2 Expression" scatter plot, and **C.** "ESR1 Expression by Pathologic Stage" violin plot.

Figure 1A shows ESR1 expression levels across all patient samples. The count data were not log-transformed for this figure, given the existing readability of the gene expression values and the general trend evident, as is. We can see that the data are right-skewed, with a small subset of samples showing very high ESR1 expression. This supports the summary statistics that suggest a positively skewed distribution in which a relatively small number of high-expression samples pull the mean upward.

Figure 1B shows the relationship between ESR1 and ERBB2 expression. We can see a high-density cluster in the upper-right region, which indicates that many samples showed relatively high ESR1 expression and relatively moderate-to-high ERBB2 expression. The fitted regression line shows a positive slope, indicating that, as ESR1 expression increases, ERBB2 expression tends to increase slightly. The correlation coefficient (r = 0.13) is small, which supports this weak association. Though the association is statistically significant (p = 3.58 x 10-6)—likely due to the large sample size—the weak correlation suggests that ESR1 and ERBB2 expression exist mostly independently such that variation in one gene does not strongly predict variation in the other.

Figure 1C shows that most patient samples showed relatively high ESR1 expression across and within the 11 pathologic stages represented in this study. We can see a few low-expression outliers at Stage I and from Stage IIB to Stage IIIC, with less such outlying expression

variation in Stage II and Stage IV. Much of the ESR1 expression clusters in the 10,000 to 100,000 count range (log10 transformed) with a wide spread within most stages. Also, we can see quite a bit of overlap in expression levels across stages. Overall, it appears that ESR1 expression is diverse both across and within pathologic stages, suggesting that ESR1 expression and pathologic stage are not strongly associated.

3.3 Heatmap Analysis

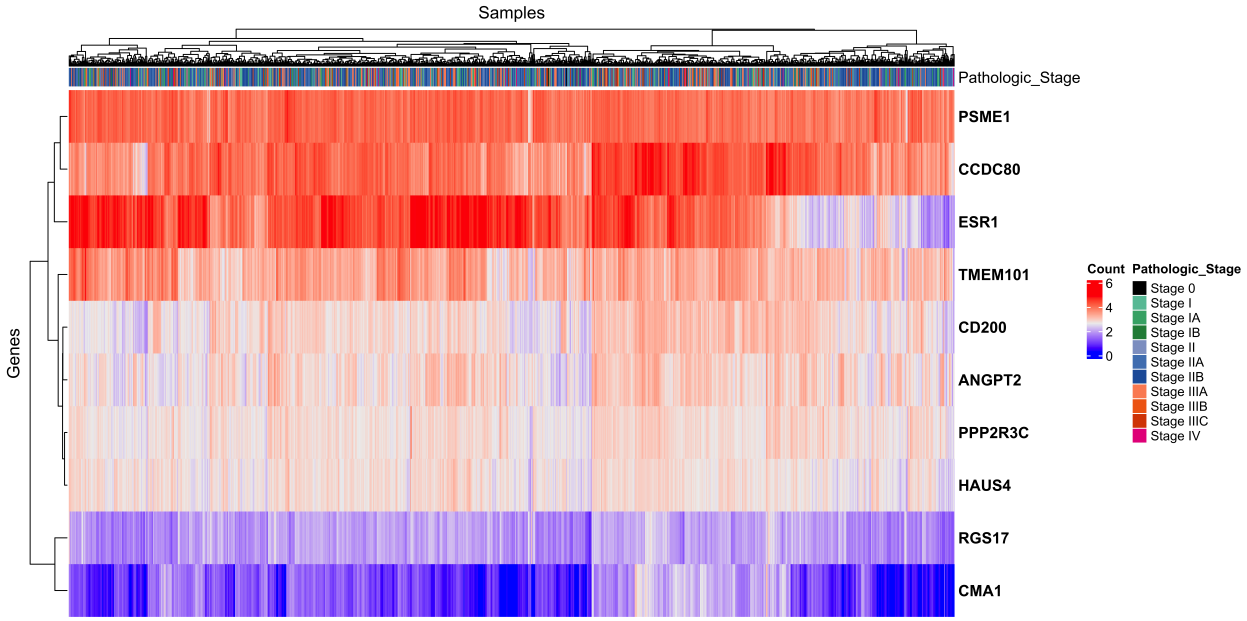


Figure 2: Heatmap of expression levels for 10 selected genes across samples, annotated by pathologic stage.

Figure 2 shows a heatmap of expression levels for 10 selected genes across all patient samples, with samples annotated by pathologic stage. The pathologic stage annotation bar shows no obvious clustering across gene expression clusters, indicating that pathologic stage is not strongly associated with the overall expression patterns of these 10 genes. In the left dendrogram, we can see that several genes appear to exhibit similar gene expression patterns regardless of pathologic stage. Overall, this heatmap suggests that pathologic stage and gene expression are not strongly associated across these 10 genes.

3.4 UMAP Analysis

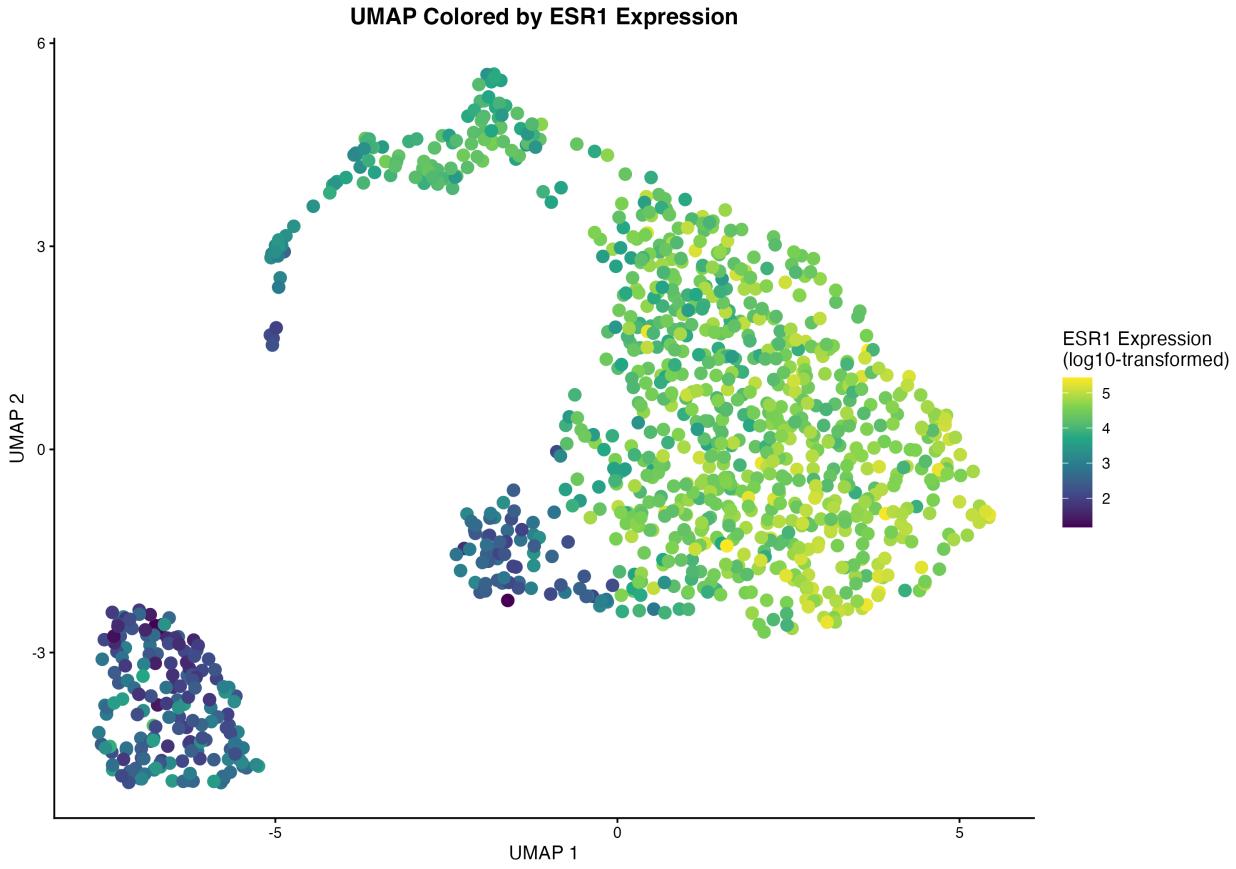


Figure 3: UMAP of overall gene expression across samples colored by ESR1 expression.

Figure 3 shows a Uniform Manifold Approximation Projection (UMAP) of overall gene expression across samples, colored by ESR1 expression. UMAP reduces the gene expression data into two dimensions such that sample proximity indicates similar overall gene expression profiles, while samples that are further apart suggest less similar overall expression patterns. Based upon this UMAP, we can see a large group of samples that appears to exhibit relatively high ESR1 expression (i.e., the lighter hues), while the lower-expression samples appear to split into two groups, one group distinctly separate from the rest (i.e., bottom left). As such, ESR1 expression is not uniformly distributed across clusters, with lower-expression samples concentrated in two clusters and higher-expression samples concentrated in one large cluster. This suggests that samples with similar ESR1 expression levels tend to exhibit similar overall gene expression profiles.

References

[1] Cheng, X. (2024). *A comprehensive review of HER2 in cancer biology and therapeutics*. Genes, 15(7), 903. https://doi.org/10.3390/genes15070903

[2] Herzog, S. K., & Fuqua, S. A. W. (2022). *ESR1 mutations and therapeutic resistance in metastatic breast cancer: Progress and remaining challenges*. British Journal of Cancer, 126(2), 174–186. https://doi.org/10.1038/s41416-021-01564-x

[3] National Cancer Institute. (n.d.). *TCGA-BRCA: Breast invasive carcinoma*. Genomic Data Commons. https://portal.gdc.cancer.gov/projects/TCGA-BRCA